

Math261 Midterm Exam 1

Exam 1 covers the following topics, and the corresponding practice problems are listed.

1. Dot product/cross product
 - Computation (Practice Q1, 3; HW1: 1-4)
 - Properties (Q5, 6; HW1: #6, 9)
 - Determine perpendicular or parallel vectors (Q4)
2. Parametric equations for lines
 - Finding parametric equations (Q7, Q8; HW3: 10-13)
 - Lines Intersection (Q9; HW3: 14-15)
3. Parametric equations of curves
 - Line segments (Q19; HW1 Q18)
 - Circles (Q15, 16; HW1: 16)
 - Ellipses (Q17,18, HW1: 17)
4. Tangent vectors to parametric curves (Q20; HW1: 20-22)
5. Equation of planes
 - Finding equation of a plane given three points on the plane, or one point and a normal vector; (Q10,11,12; HW2: #2-6)
 - Finding parametric equations for the line of intersection of two planes (Q13,14; HW2:11-12)
 - Finding the intersection point(s) of a line and a plane (Q7, Q8; HW2: 7-9)
6. Partial derivative
 - Computing first-order, and second-order partial derivatives (Q21-25; HW2:13-15)
 - Chain Rule (Q41-46; HW2: 16-21)
 - Determine the sign of a partial derivative by visualizing a surface graph (Q48-49)
7. Directional derivative and gradient vector
 - Computing the gradient vector (HW3: #2-3); finding the directions of steepest increase or decrease (Q29, 30, 31; HW3: #4-8); draw the gradient vector relative to level curves (perpendicular to the curve, pointing toward increasing values) (Q49-50a,b)
 - Computing directional derivative in a given direction (Q26; HW3: #9-10, 13-15); Using its sign to determine whether the function is increasing or decreasing (Q27, 28; HW3: 11-12), and confirm that conclusion against the graph (Q49-50c)
 - Finding the equation of the tangent plane to a surface (Q32-35; HW3: 22-25)
8. Critical points and local extreme values
 - Finding critical points, and classify the type for each point (Q36-40; HW3: 16-25)

MATH 261: Exam 1 Practice Problems

1. Let $\mathbf{u} = 1\mathbf{i} + 1\mathbf{j} + 1\mathbf{k}$ and $\mathbf{v} = 0\mathbf{i} + -1\mathbf{j} + 2\mathbf{k}$. Compute $3\mathbf{u} + \mathbf{v}$.
2. Let $\mathbf{v} = 2\mathbf{i} + 1\mathbf{j} + -2\mathbf{k}$. Find a unit vector that points in the same direction as $\mathbf{v}$.
3. Let $\mathbf{u} = \langle 1, -1, 1 \rangle$ and $\mathbf{v} = \langle 2, 2, 0 \rangle$. Compute the dot product $\mathbf{u} \cdot \mathbf{v}$ and the cross product $\mathbf{u} \times \mathbf{v}$.
4. Suppose that $\mathbf{u}$ and $\mathbf{v}$ are two nonzero vectors. What would the equation $\mathbf{u} \cdot \mathbf{v} = 0$ tell you about $\mathbf{u}$ and $\mathbf{v}$ geometrically? What would the equation $\mathbf{u} \times \mathbf{v} = 0$ tell you about $\mathbf{u}$ and $\mathbf{v}$ geometrically?
5. If $\mathbf{u} \cdot \mathbf{v} = 2$, what must $3\mathbf{u} \cdot 2\mathbf{v} + \mathbf{v} \cdot 4\mathbf{u}$ equal?
6. If $\mathbf{u} \times \mathbf{v} = 2\mathbf{j} - \mathbf{k}$, what must $2\mathbf{v} \times \mathbf{u}$ equal?
7. Find the parametric equations of the line that goes through the point $(3, 1, -1)$ and is parallel to the vector $\langle 2, 4, 1 \rangle$. At what point or points does this line intersect the plane $x - y + 2z = 0$, if any?
8. Find the parametric equations of the line that goes through the points $(1, 2, 2)$ and $(3, 3, 0)$. At what point or points, if any, does this line intersect the plane $x + z = 1$, if any?
9. At which point or points, if any, does the line through the points $(2, 3, 1)$ and $(3, 4, 1)$ intersect the line which contains the point $(0, 2, 0)$ and is parallel to the vector $\langle 1, 0, 1 \rangle$?
10. Find an equation for the plane which contains the point $(2, 3, 5)$ and whose normal line is in the direction of $\langle 1, 1, -1 \rangle$.
11. Find an equation for the plane which contains the points $(1, 1, 2)$, $(4, 5, -2)$, and $(0, 0, 1)$.
12. Find an equation for the plane which contains the points $(2, 1, 0)$, $(1, 0, 2)$, and $(0, -1, -4)$.
13. Find a parametric equation for the line on which the planes $x + y - z = 0$ and $x + z = 3$ meet.
14. Find a parametric equation for the line on which the planes $x + 2y + z = 0$ and $x - y + 2z = 4$ meet.
15. Find a parametric equation for the circle $x^2 + y^2 = 1$.
16. Find a parametric equation for the circle $x^2 + y^2 = 16$.
17. Find a parametric equation for the ellipse $x^2 + \frac{y^2}{81} = 1$.
18. Find a parametric equation for the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 100$.
19. Find a parametric equation for the triangle with vertices at $(0, 0)$, $(2, 0)$, and $(0, 5)$.

20. Consider the curve parameterized by $\mathbf{r}(t) = (t^3, e^t + 1)$ for $0 \leq t \leq 1$.
- What are the starting and ending points of this curve?
 - If this curve actually represented the position of an object, what would be the velocity vector of the object?
 - Compute an expression for the unit tangent vector, $\mathbf{T}(t)$.
21. Suppose $f(x, y) = \sin(x^2y)$. Compute the partial derivatives $f_x(x, y)$ and $f_y(x, y)$.
22. Suppose $f(x, y) = x \tan(x + y)$. Compute the partial derivatives $f_x(x, y)$ and $f_y(x, y)$.
23. Suppose $f(x, y) = \ln(100 + x^2 + y^4)$. Compute the partial derivatives $f_x(x, y)$ and $f_y(x, y)$.
24. Suppose $f(x, y) = x^3y + y^2x^2 + 2x - y + 12$. Compute the partial derivatives $f_x(x, y)$, $f_y(x, y)$, $f_{xx}(x, y)$, $f_{yy}(x, y)$ and $f_{xy}(x, y)$.
25. Suppose $f(x, y, z) = x + x^2y^2 + x^2y^2z^2$. Compute the partial derivatives $f_x(x, y, z)$, $f_y(x, y, z)$, and $f_z(x, y, z)$.
26. Suppose $\nabla f = 2\mathbf{i} - 3\mathbf{j} + \mathbf{k}$ at $(0, 0, 0)$. What is the directional derivative of this function at the origin, $D_{\mathbf{u}}f(0, 0, 0)$, where $\mathbf{u}$ is a vector which points in the direction of $\mathbf{u} = \langle 1, 1, -1 \rangle$?
27. Is the function $f(x, y) = x^2y^2 + x^4 + e^y$ increasing, decreasing or instantaneously constant at the point $(-1, 0)$ in the direction $\langle 1, 1 \rangle$?
28. Is the function $f(x, y, z) = ze^{x^2y}$ increasing, decreasing or instantaneously constant at the point $(1, 1, 1)$ in the direction $\langle 3, -1, -2 \rangle$?
29. Suppose $g(x, y) = ye^x$.
- Compute the gradient $\nabla g(x, y)$.
 - What is the direction of greatest increase for this function at $(0, 0)$ (expressed as a unit vector)?
 - Compute the directional derivative $D_{\mathbf{u}}f(x, y)$ at the point $(0, 0)$, where $\mathbf{u}$ points in the direction of $\langle 1, -2 \rangle$.
 - Compute the directional derivative $D_{\mathbf{u}}f(x, y)$ at the point $(0, 0)$, where $\mathbf{u}$ points in the direction of $\langle 2, -4 \rangle$.
 - Compute the directional derivative $D_{\mathbf{u}}f(x, y)$ at the point $(0, 0)$, where $\mathbf{u}$ points in the direction of $\langle 5, 0 \rangle$.
30. Suppose $g(x, y) = e^{-x^2y^2}$.
- Compute the gradient $\nabla g(x, y)$.
 - What is the direction of greatest increase for this function at $(1, -1)$ (expressed as a unit vector)?
 - Compute the directional derivative $D_{\mathbf{u}}f(x, y)$ at the point $(1, -1)$, where $\mathbf{u}$ points in the direction of $\langle 2, 1 \rangle$.
31. Suppose $h(x, y, z) = x^2y + y^2x + z^3$.
- Compute the gradient $\nabla h(x, y, z)$.

- b) What is the direction of greatest increase for this function at $(1, 1, 1)$ (expressed as a unit vector)?
- c) Compute the directional derivative $D_{\mathbf{u}}f(x, y)$ at the point $(1, 1, 1)$, where $\mathbf{u}$ points in the direction of $\langle 1, 2, 1 \rangle$.
32. Find an equation of tangent plane to the surface $x^2 + y^2 + z^2 = 3$ at the point $(1, 1, 1)$.
33. Find an equation of tangent plane to the surface $\cos(\pi x) - x^2y + e^{xz} + yz = 4$ at the point $(0, 1, 2)$.
34. Find an equation of tangent plane to the surface $z = \ln(x^2 + y^2)$ at the point $(1, 0, 0)$.
35. Find an equation of tangent plane to the surface $z = \sqrt{y - x}$ at the point $(1, 2, 1)$.
36. Is $(1, 1)$ a critical point of the function $f(x, y) = x^4 - y^2x^2 + x^2 - y^2$? If so, classify it as a local maximum, minimum, or saddle point. If not, explain why not.
37. Suppose $f(x, y)$ is a function so that $f_x(1, 2) = 0$, $f_y(1, 2) = 0$, and suppose that $f_{xx}(1, 2) = 3$, $f_{yy}(1, 2) = 2$, and $f_{xy}(1, 2) = 1$. Is $(1, 2)$ a local maximum, local minimum, or saddle point for this function?
38. Suppose $f(x, y)$ is a function so that $f_x(1, 2) = 0$, $f_y(1, 2) = 0$, and suppose that $f_{xx}(1, 2) = 5$, $f_{yy}(1, 2) = 3$, and $f_{xy}(1, 2) = 4$. Is $(1, 2)$ a local maximum, local minimum, or saddle point for this function?
39. Consider a function $f(x, y) = xy - x^2 - y^2 - 2x - 2y + 4$,
- Find critical point(s).
 - Use the second derivative test to determine if each critical point corresponds to a local maximum, local minimum or saddle point? If it corresponds to a local maximum or minimum, find its value.
40. Consider a function $f(x, y) = -3x^2 + 3y^2 - 2y^3 + 6xy$,
- Find critical point(s).
 - Use the second derivative test to determine if each critical point corresponds to a local maximum, local minimum or saddle point? If it corresponds to a local maximum or minimum, find its value.
41. Use the Chain Rule to evaluate $\frac{df}{dt}$ for the following function.
- $$f(x, y) = x^2y + 3xy^4, \quad x(t) = \sin(2t), \quad y(t) = \cos(t)$$
42. Use the Chain Rule to evaluate $\frac{df}{dt}$ for the following function.
- $$f(x, y) = e^{x^2 - y}, \quad x = \sin(4t), \quad y = t^3 - 9$$
43. Use the Chain Rule to evaluate $\frac{df}{dt}$ for the following function.
- $$f(x, y, z) = z - \sin(xy), \quad x = t, \quad y = \ln t, \quad z = e^{t-1}$$

44. Use the Chain Rule to evaluate $\frac{df}{dt}$ for the following function.

$$f(x, y, z) = x^4 - 4xy^2 + z^3, \quad x = \sqrt{t}, \quad y = e^{2t}, \quad z = \frac{1}{t}$$

45. Use the Chain Rule to evaluate $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$ for the following function.

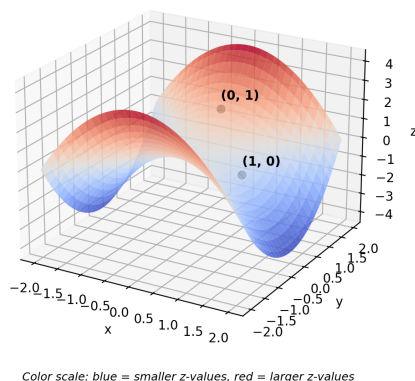
$$z = x^2y + x, \quad x = s^2 + 3t, \quad y = 2s - t$$

46. Use the Chain Rule to evaluate $\frac{\partial w}{\partial s}$ and $\frac{\partial w}{\partial t}$ for the following function.

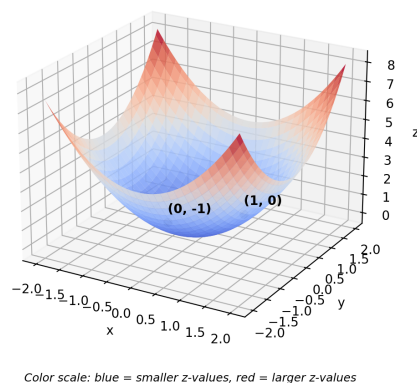
$$w = xy + z^2, \quad \text{where } x = t^2e^s, \quad y = t \cos(s)$$

47. (*Conceptual / Visualization*)

The surface $z = f(x, y) = y^2 - x^2$ is shown below, with two points marked on the surface.



- (a) At the point $(1, 0)$, holding $y = 0$ fixed, is the cross-section of the surface increasing or decreasing as x increases? What does this tell you about the sign of $\frac{\partial f}{\partial x}$ at $(1, 0)$?
- (b) At the point $(0, 1)$, use the graph to determine the sign of $\frac{\partial f}{\partial y}$.
- (c) Based on the shape of the surface, classify the critical point at the origin as a local maximum, local minimum, or saddle point. Explain your reasoning using the graph alone (no computation required)
48. (*Conceptual / Visualization*)
- The surface $z = f(x, y) = x^2 + y^2$ is shown below, with two points marked on the surface.



- At the point $(1, 0)$, holding $y = 0$ fixed, is the cross-section increasing or decreasing as x increases? What does this tell you about the sign of $\frac{\partial f}{\partial x}$ at $(1, 0)$?
- At the point $(0, -1)$, use the graph to determine the sign of $\frac{\partial f}{\partial y}$.
- Based on the shape of the surface, classify the critical point at the origin as a local maximum, local minimum, or saddle point. Explain your reasoning using the graph alone (no computation required)

49. (Conceptual / Visualization)

Consider the surface $z = f(x, y) = 18 - x^2 - 3y^2$, shown in Figure (a), with the point $(2, 1, 11)$ marked.

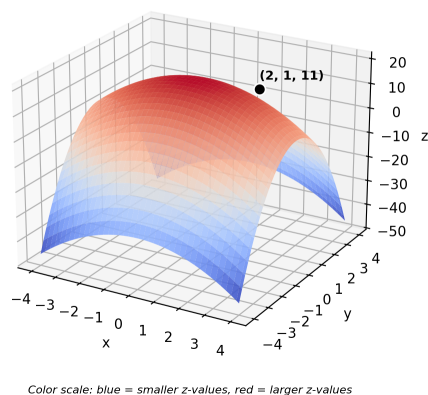


Figure (a)

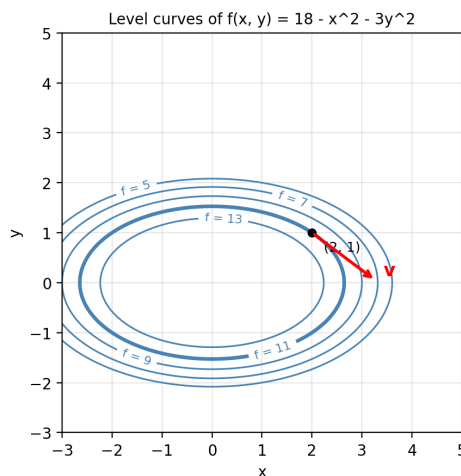


Figure (b)

- Compute ∇f at $(2, 1)$.
- The level curve of f through $(2, 1)$ (i.e., $f(x, y) = 11$) is shown in Figure (b), along with two neighboring level curves on either side for context. On the graph, draw the gradient vector ∇f at $(2, 1)$, in the correct direction (you do not need to scale it precisely). Briefly explain how you knew which way it should point.

- (c) The red arrow in Figure (b) represents the direction vector $\mathbf{v} = \langle 4, -3 \rangle$. Compute the directional derivative of f at $(2, 1)$ in that direction, and determine whether f is increasing or decreasing. Use the graph to confirm your conclusion.

50. (Conceptual / Visualization)

Consider the surface $z = f(x, y) = x^2 - y^2$, shown in Figure (a), with the point $(1, 2, -3)$ marked.

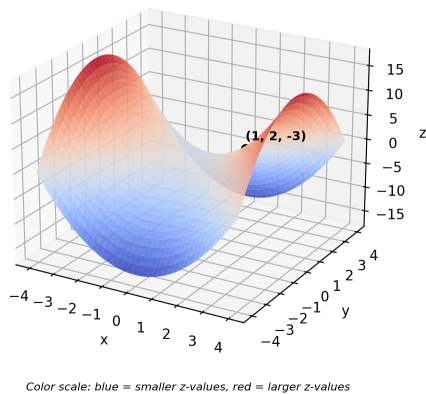


Figure (a)

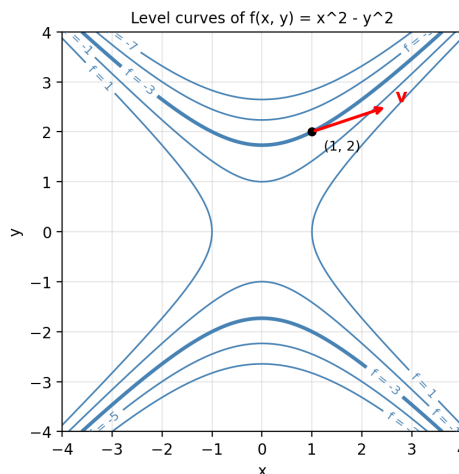


Figure (b)

- (a) Compute ∇f at $(1, 2)$.
- (b) The level curve of f through $(1, 2)$ (i.e., $f(x, y) = -3$) is shown in Figure (b), along with two neighboring level curves on either side. On the graph, draw the gradient vector ∇f at $(1, 2)$, in the correct direction (you do not need to scale it precisely). Briefly explain how you knew which way it should point.
- (c) The red arrow in Figure (b) represents the direction vector $\mathbf{v} = \langle 3, 1 \rangle$. Compute the directional derivative of f at $(1, 2)$ in that direction, and determine whether f is increasing or decreasing. Use the graph to confirm your conclusion.

Exam 1 Practice Problem Solutions

NOTE: For *many* of these problems, there are multiple possible mathematically equivalent correct answers.

1. $\langle 3, 2, 5 \rangle$
2. $\frac{1}{\sqrt{9}} \langle 2, 1, -2 \rangle$
3. Dot product is 0, cross product is $-2\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$.
4. The first one tells you that they are orthogonal (at a ninety degree angle in their common plane) to each other. The second one tells you that they are parallel to one another.
5. 20
6. $-4\mathbf{j} + 2\mathbf{k}$
7. The line is entirely on the plane.
8. It does not.
9. $(1, 2, 1)$
10. $(x - 2) + (y - 3) - (z - 5) = 0$
11. $-8(x - 1) + 7(y - 1) + (z - 2) = 0$
12. $x - y = 1$
13. $x(t) = 1 + t, y(t) = 1 - 2t, z(t) = 2 - t$ (other, equivalent answers possible)
14. $x(t) = 1 + 5t, y(t) = -1 - t, z(t) = 1 - 3t$ (other, equivalent answers possible)
15. $x(t) = \cos(t), y(t) = \sin(t), \quad 0 \leq t \leq 2\pi$
16. $x(t) = 4\cos(t), y(t) = 4\sin(t), \quad 0 \leq t \leq 2\pi$
17. $x(t) = \cos(t), y(t) = 9\sin(t), \quad 0 \leq t \leq 2\pi$
18. $x(t) = 50\cos(t), y(t) = 30\sin(t), \quad 0 \leq t \leq 2\pi$
19.
$$(x(t), y(t)) = \begin{cases} (2t, 0), & 0 \leq t \leq 1 \\ (2 - t, 5(t - 1)), & 1 \leq t \leq 2 \\ (0, 5(3 - t)), & 2 \leq t \leq 3 \end{cases}$$
20. a) $(0, 2)$ and $(1, e + 1)$
b) $\langle 3t^2, e^t \rangle$
c) $\left\langle \frac{3t^2}{\sqrt{9t^4 + e^{2t}}}, \frac{e^t}{\sqrt{9t^4 + e^{2t}}} \right\rangle$
21. $f_x = 2xy \cos(x^2y), \quad f_y = x^2 \cos(x^2y)$

22. $f_x = \tan(x+y) + x \sec^2(x+y)$, $f_y = x \sec^2(x+y)$
23. $f_x = \frac{2x}{100+x^2+y^4}$, $f_y = \frac{4y^3}{100+x^2+y^4}$
24. $f_x = 3x^2y + 2xy^2 + 2$, $f_y = x^3 + 2x^2y - 1$, $f_{xx} = 6xy + 2y^2$, $f_{yy} = 2x^2$, $f_{xy} = 3x^2 + 4xy$
25. $f_x = 1 + 2xy^2 + 2xy^2z^2$, $f_y = 2x^2y + 2x^2yz^2$, $f_z = 2x^2y^2z$
26. $\frac{-2}{\sqrt{3}}$
27. Decreasing (directional derivative is $-3/\sqrt{2}$)
28. It is instantaneously constant.
29. a) $\langle ye^x, e^x \rangle$
 b) $\langle 0, 1 \rangle$
 c) $\frac{-2}{\sqrt{5}}$
 d) $\frac{-2}{\sqrt{5}}$
 e) 0
30. a) $\langle -2xy^2e^{-x^2y^2}, -2x^2ye^{-x^2y^2} \rangle$
 b) $\left\langle \frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle$
 c) $\frac{-2}{\sqrt{5}e}$
31. a) $\langle 2xy + y^2, x^2 + 2xy, 3z^2 \rangle$
 b) $\left\langle \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$
 c) $12/\sqrt{6}$
32. $x + y + z = 3$
33. $2x + 2y + z = 4$
34. $2x - z = 2$
35. $x - y + 2z = 1$
36. It is not a critical point; $f_y \neq 0$.
37. Local minimum
38. Saddle point
39. (a) $(-2, -2)$ is the only critical point.
 (b) $(-2, -2)$ corresponds to a local maximum; the local maximum value of f at this point is $f(-2, -2) = 8$.

40. (a) $(0, 0)$ and $(2, 2)$ are two critical points.
 (b) Critical point $(0, 0)$ corresponds to a saddle point; critical point $(2, 2)$ corresponds to a local maximum. The local maximum value of f is $f(2, 2) = 8$.

41. $\frac{df}{dt} = (2xy + 3y^4)(2\cos(2t)) + (x^2 + 12xy^3)(-\sin(t))$

42. $\frac{df}{dt} = 2xe^{x^2-y}(4\cos(4t)) - e^{x^2-y}(3t^2)$

43. $\frac{df}{dt} = -y\cos(xy) - x\cos(xy)\frac{1}{t} + e^{t-1}$

44. $\frac{df}{dt} = (4x^3 - 4y^2)\left(\frac{1}{2}t^{-1/2}\right) - 16xye^{2t} - 3z^2t^{-2}$

45. $\frac{\partial z}{\partial s} = 4xys + 2s + 2x^2; \quad \frac{\partial z}{\partial t} = 6xt - x^2 + 3$

46. $\frac{\partial w}{\partial t} = y(2te^s) + x\cos(s); \quad \frac{\partial w}{\partial s} = y(t^2e^s) + x(-t\sin(s))$

47. $f(x, y) = y^2 - x^2, \quad f_x = -2x, \quad f_y = 2y.$

- (a) At $(1, 0)$, holding $y = 0$ fixed: $f(x, 0) = -x^2$, which is **decreasing** as x increases. This tells us $\frac{\partial f}{\partial x} < 0$ at $(1, 0)$.

$$f_x(1, 0) = -2(1) = -2 < 0. \checkmark$$

Graph confirmation: near $(1, 0)$, moving in the direction of increasing x , the surface becomes bluer (smaller z) — consistent with f decreasing.

- (b) At $(0, 1)$: $f_x(0, 1) = -2(0) = 0$, and $f_y(0, 1) = 2(1) = 2 > 0$. So $\frac{\partial f}{\partial y} > 0$ at this point.

Graph confirmation: near $(0, 1)$, moving in the direction of increasing y , the surface rises toward the red (larger- z) region — consistent with $f_y > 0$.

- (c) Critical point: solving $f_x = -2x = 0$ and $f_y = 2y = 0$ gives $(0, 0)$.

$$f_{xx} = -2, \quad f_{yy} = 2, \quad f_{xy} = 0, \quad D = f_{xx}f_{yy} - f_{xy}^2 = (-2)(2) - 0 = -4 < 0.$$

Since $D < 0$, $(0, 0)$ is a **saddle point**. This matches the graph: the surface rises along the y -axis and falls along the x -axis near the origin.

48. $f(x, y) = x^2 + y^2, \quad f_x = 2x, \quad f_y = 2y.$

- (a) At $(1, 0)$, holding $y = 0$ fixed: $f(x, 0) = x^2$, which is **increasing** as x increases. This tells us $\frac{\partial f}{\partial x} > 0$ at $(1, 0)$.

$$f_x(1, 0) = 2(1) = 2 > 0. \checkmark$$

Graph confirmation: near $(1, 0)$, moving in the direction of increasing x , the surface becomes redder (larger z) — consistent with f increasing.

- (b) At $(0, -1)$: $f_x(0, -1) = 2(0) = 0$, and $f_y(0, -1) = 2(-1) = -2 < 0$. So $\frac{\partial f}{\partial y} < 0$ at this point.

Graph confirmation: near $(0, -1)$, moving in the direction of increasing y (toward the origin), the surface descends toward the blue (smaller- z) minimum — consistent with $f_y < 0$.

- (c) Critical point: solving $f_x = 2x = 0$ and $f_y = 2y = 0$ gives $(0, 0)$.

$$f_{xx} = 2, \quad f_{yy} = 2, \quad f_{xy} = 0, \quad D = f_{xx}f_{yy} - f_{xy}^2 = (2)(2) - 0 = 4 > 0, \quad f_{xx} > 0.$$

Since $D > 0$ and $f_{xx} > 0$, $(0, 0)$ is a **local minimum**. This matches the graph: the surface is a bowl opening upward, with its lowest point at the origin.

49. $f(x, y) = 18 - x^2 - 3y^2$, $f_x = -2x$, $f_y = -6y$. Point: $(2, 1)$, with $\mathbf{v} = \langle 4, -3 \rangle$.

- (a)

$$\nabla f(2, 1) = \langle f_x(2, 1), f_y(2, 1) \rangle = \langle -2(2), -6(1) \rangle = \langle -4, -6 \rangle.$$

- (b) The gradient always points perpendicular to the level curve, in the direction in which f increases most rapidly. Since $f(2, 1) = 11$ and the neighboring curve $f = 13$ lies closer to the origin, $\nabla f(2, 1) = \langle -4, -6 \rangle$ should be drawn perpendicular to the $f = 11$ curve, pointing inward (toward the origin) — toward that higher-valued curve.

- (c) Direction: $\mathbf{v} = \langle 4, -3 \rangle$, $|\mathbf{v}| = 5$, so $\mathbf{u} = \langle \frac{4}{5}, -\frac{3}{5} \rangle$.

$$D_{\mathbf{u}}f(2, 1) = \nabla f(2, 1) \cdot \mathbf{u} = (-4) \left(\frac{4}{5} \right) + (-6) \left(-\frac{3}{5} \right) = \frac{-16+18}{5} = \frac{2}{5} > 0.$$

Since $D_{\mathbf{u}}f(2, 1) > 0$, f is **increasing** in the direction of $\mathbf{v}$.

Graph confirmation: the red arrow in Figure (b) points from the $f = 11$ curve toward the $f = 13$ curve — i.e., toward higher level-curve values — consistent with f increasing in that direction.

50. $f(x, y) = x^2 - y^2$, $f_x = 2x$, $f_y = -2y$. Point: $(1, 2)$, with $\mathbf{v} = \langle 3, 1 \rangle$.

- (a)

$$\nabla f(1, 2) = \langle f_x(1, 2), f_y(1, 2) \rangle = \langle 2(1), -2(2) \rangle = \langle 2, -4 \rangle.$$

- (b) The gradient always points perpendicular to the level curve, in the direction in which f increases most rapidly. Since $f(1, 2) = -3$ and the neighboring curve $f = -1$ lies closer to the x -axis (toward larger, less negative values), $\nabla f(1, 2) = \langle 2, -4 \rangle$ should be drawn perpendicular to the $f = -3$ curve, pointing toward that higher-valued curve.

- (c) Direction: $\mathbf{v} = \langle 3, 1 \rangle$, $|\mathbf{v}| = \sqrt{10}$, so $\mathbf{u} = \left\langle \frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}} \right\rangle$.

$$D_{\mathbf{u}}f(1, 2) = \nabla f(1, 2) \cdot \mathbf{u} = (2) \left(\frac{3}{\sqrt{10}} \right) + (-4) \left(\frac{1}{\sqrt{10}} \right) = \frac{6-4}{\sqrt{10}} = \frac{2}{\sqrt{10}} > 0.$$

Since $D_{\mathbf{u}}f(1, 2) > 0$, f is **increasing** in the direction of $\mathbf{v}$.

Graph confirmation: the red arrow in Figure (b) points from the $f = -3$ curve toward the $f = -1$ curve — i.e., toward higher (less negative) level-curve values — consistent with f increasing in that direction.